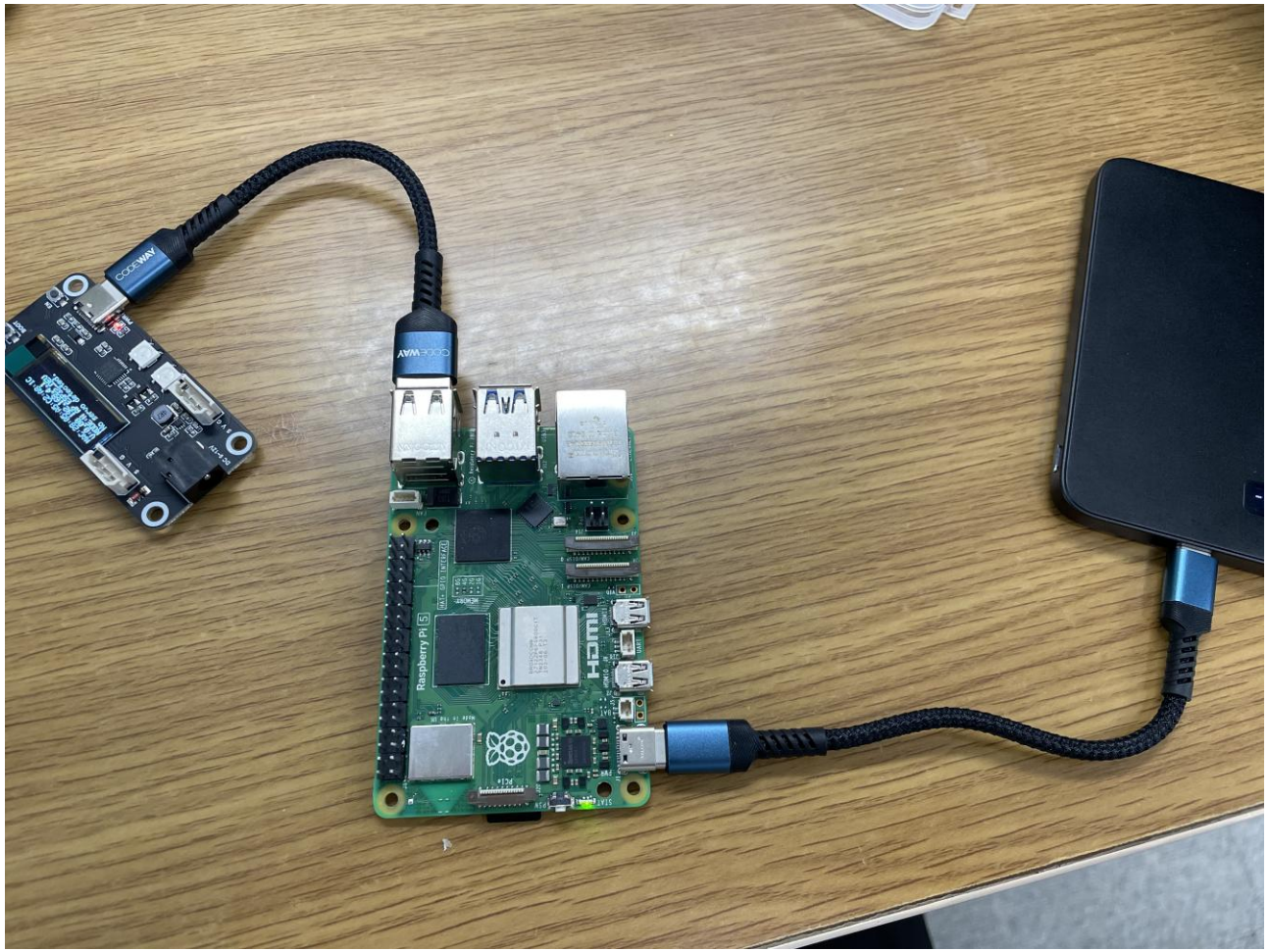
1. Setup ESP32

Make the connection like this.

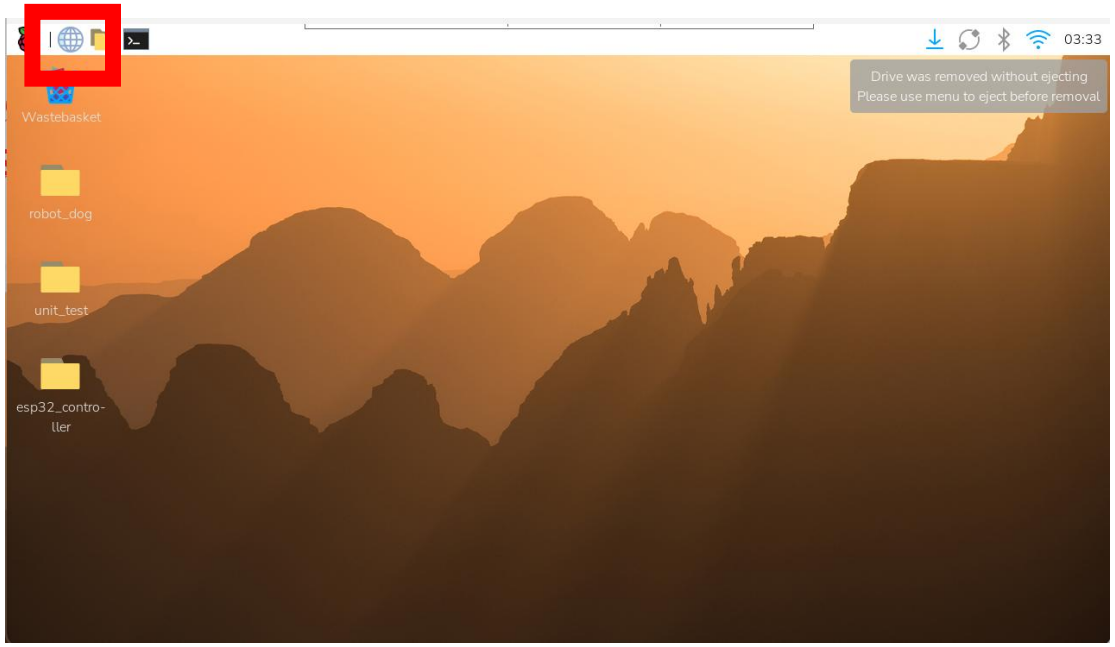


Turn on the raspberry pi

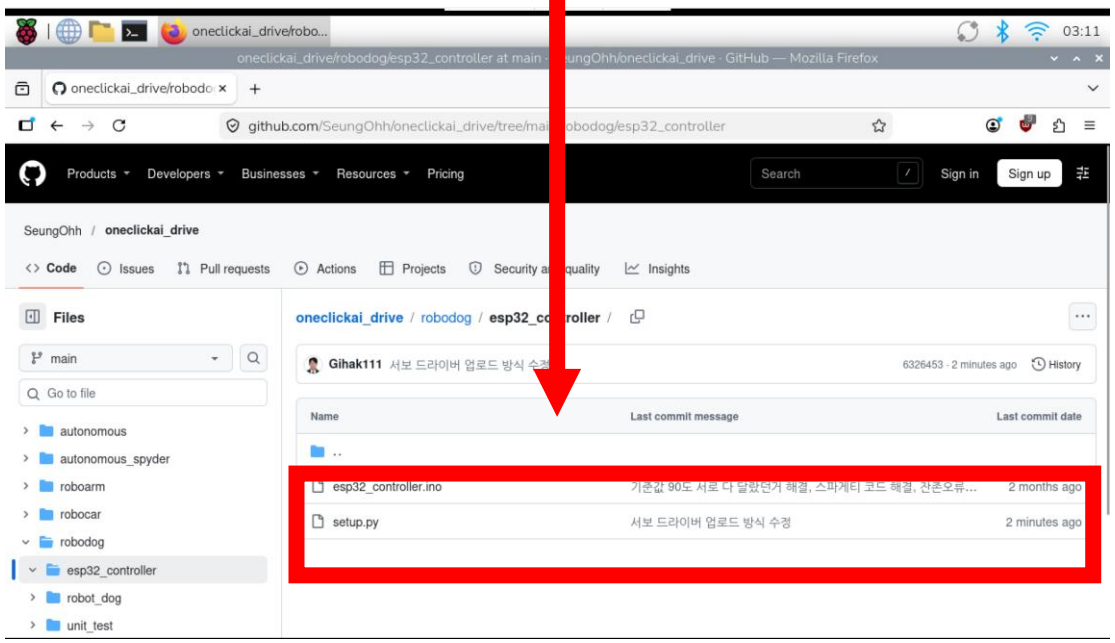
1. Setup ESP32

https://github.com/SeungOhh/oneclickai_drive/tree/main/robodog/esp32_controller in raspberry pi

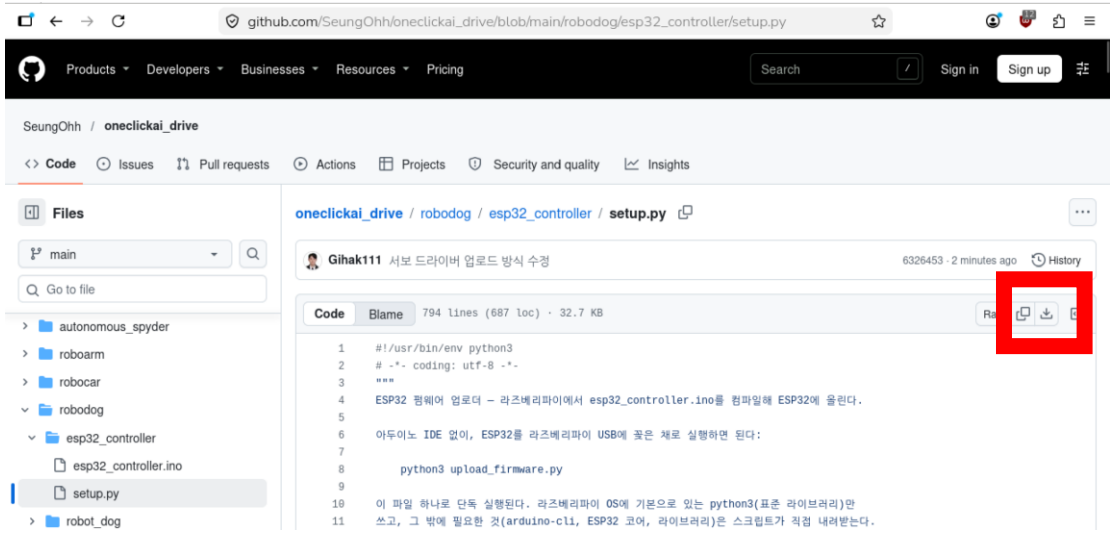
Open web first



Download this file

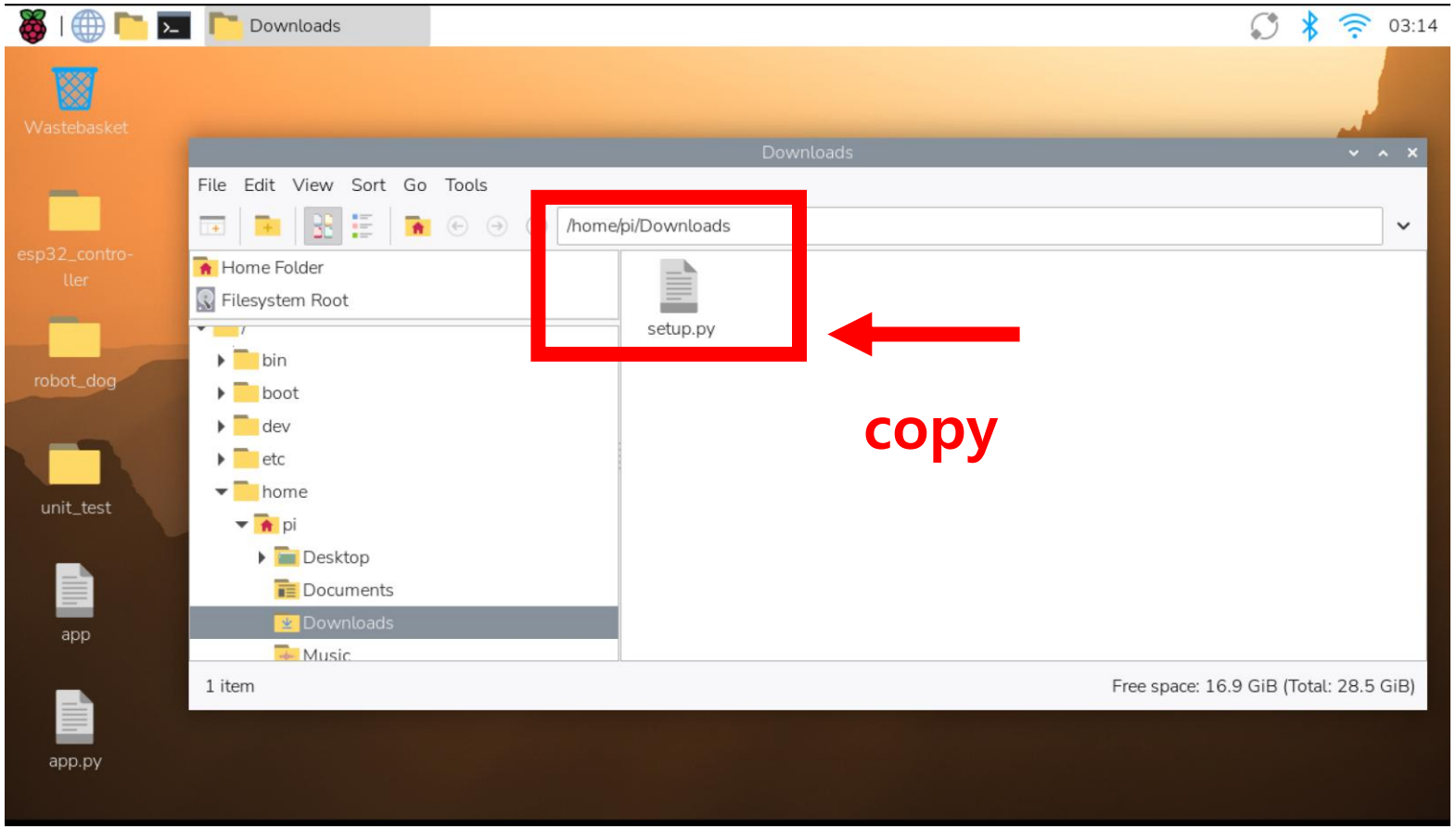


1. Setup ESP32

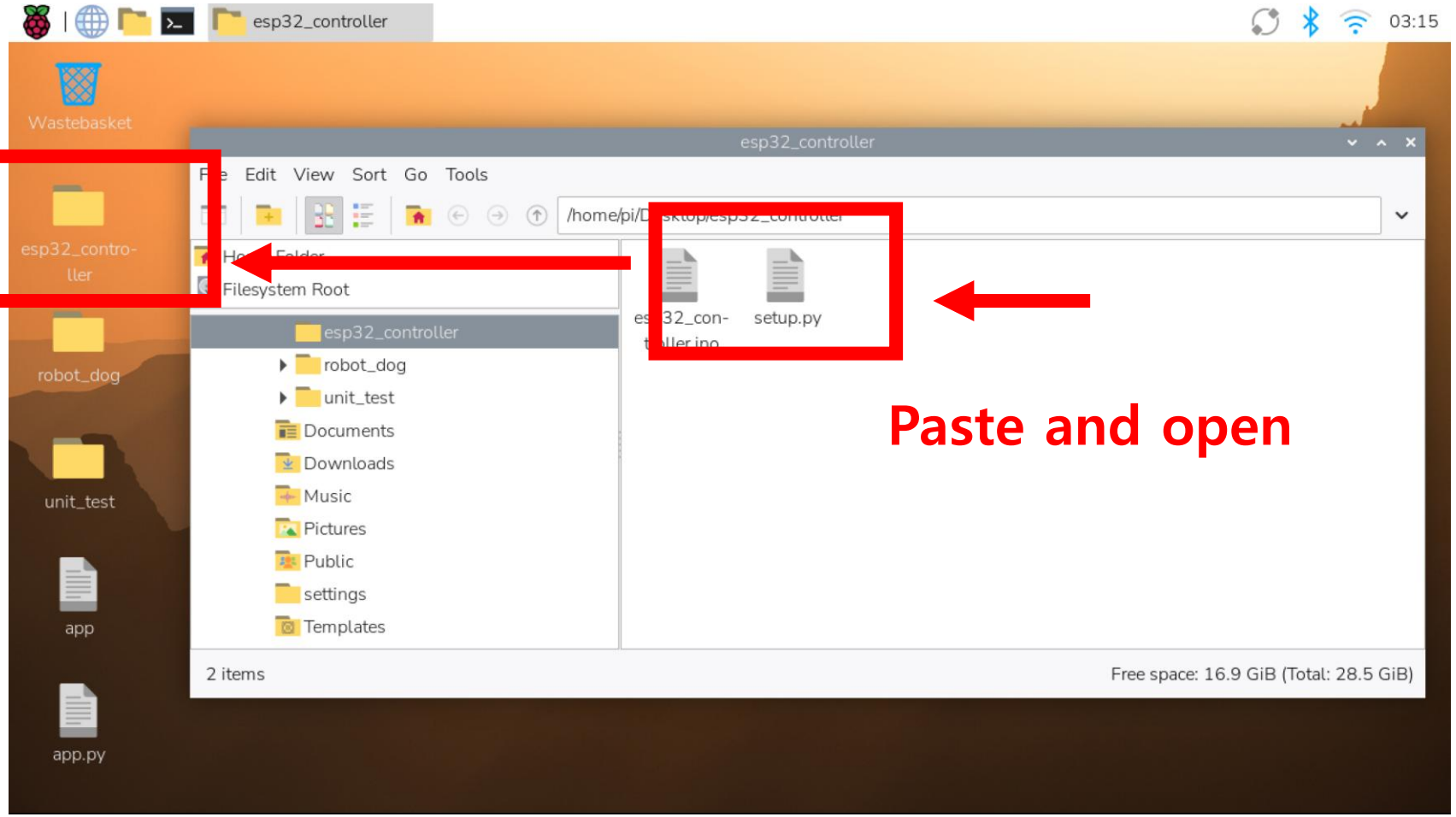


Download this file

1. Setup ESP32

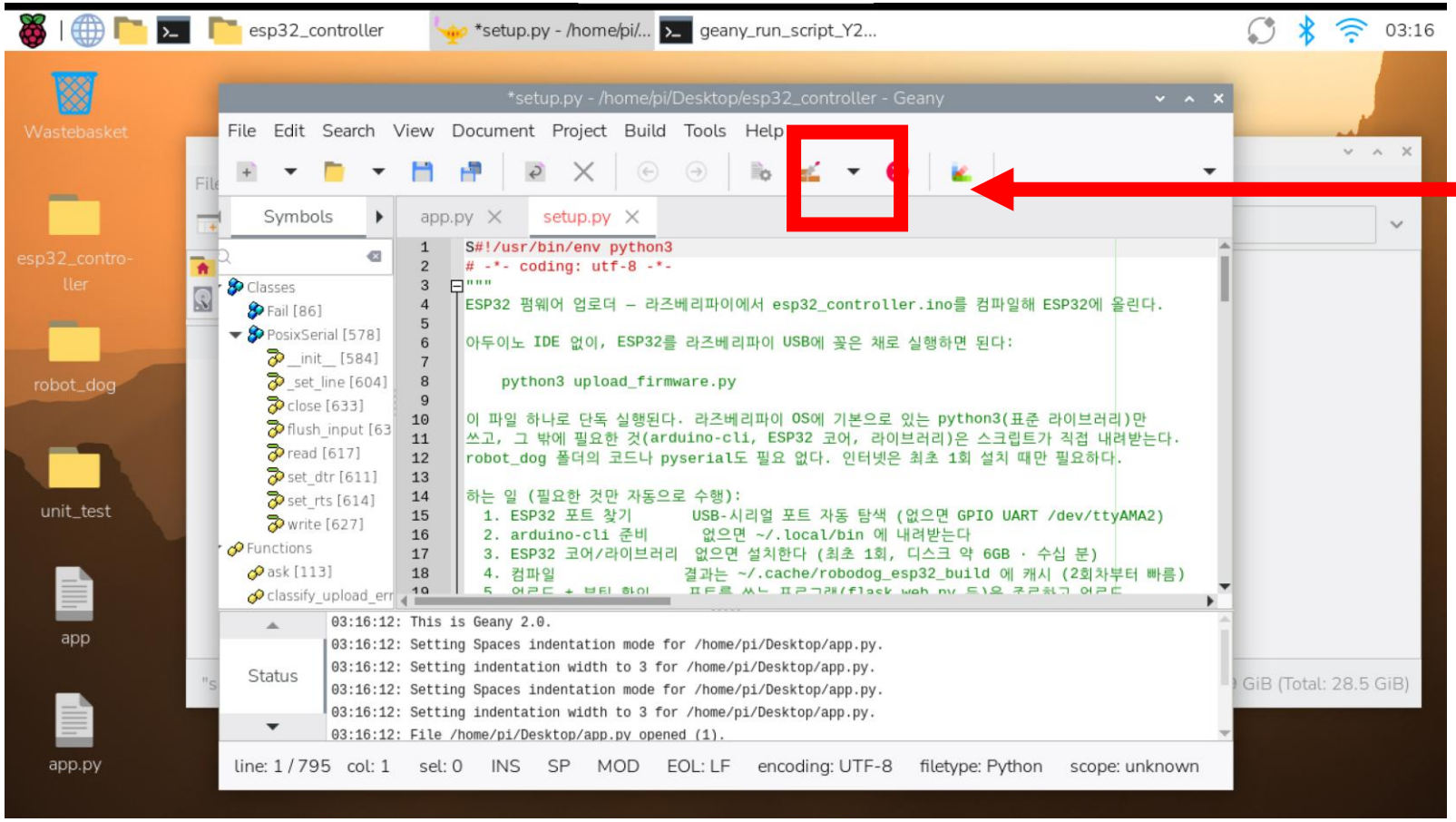


1. Setup ESP32



1. Setup ESP32

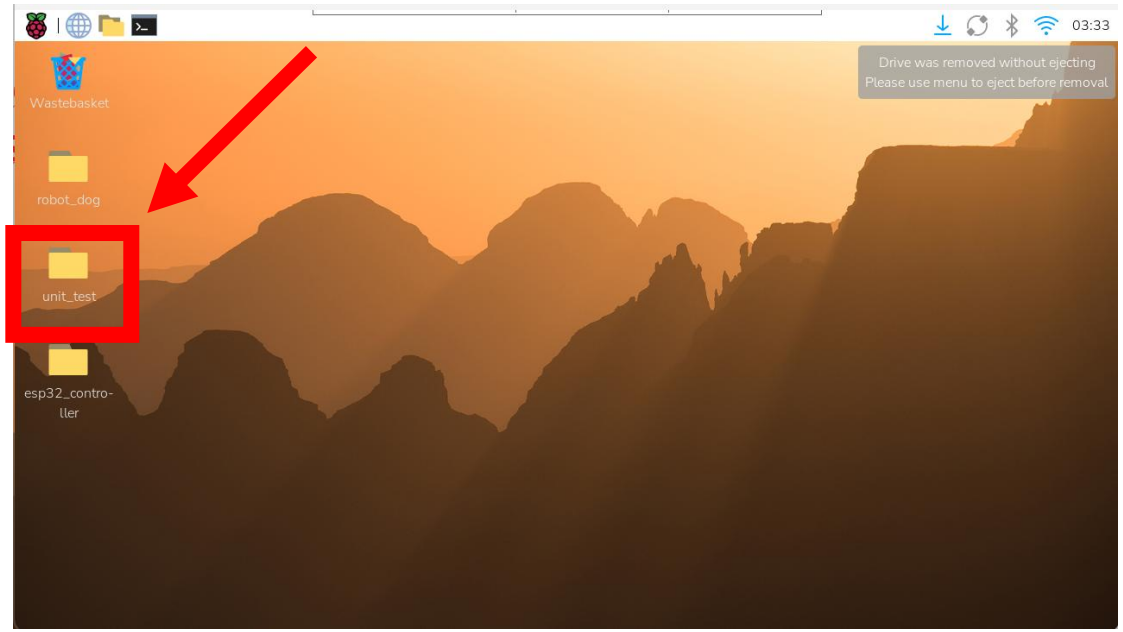
Copy downloaded file and past to esp32 folder

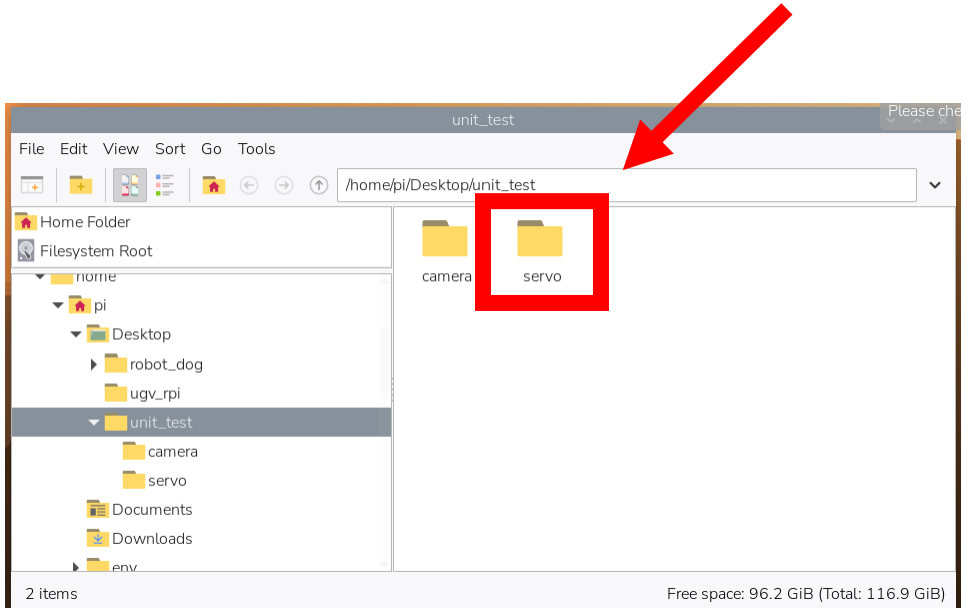


Run and wait

2. Servo test

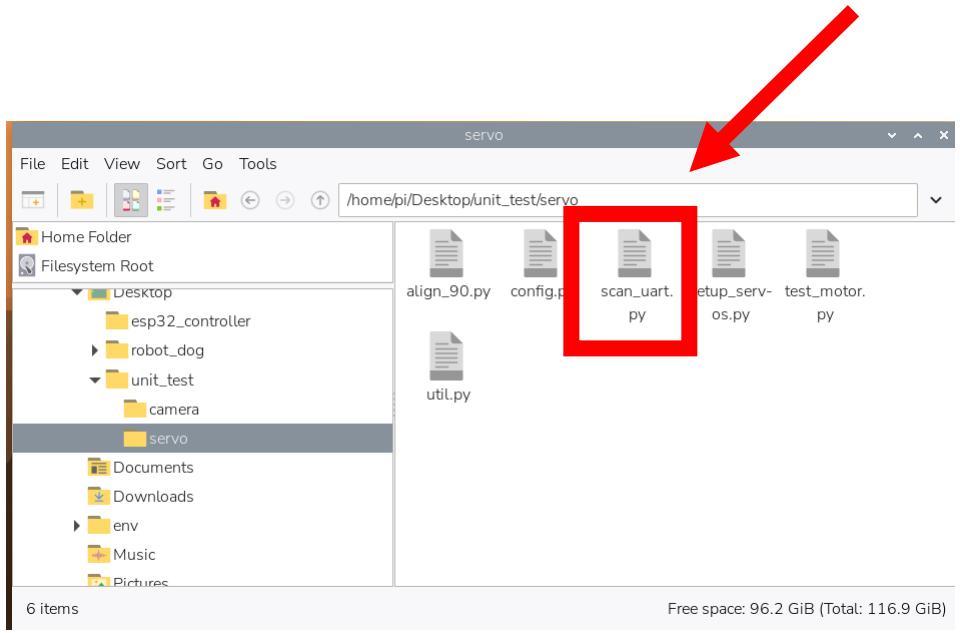
We need to disassemble and rebuild it for servo calibration because it wasn't done during the last assembly.

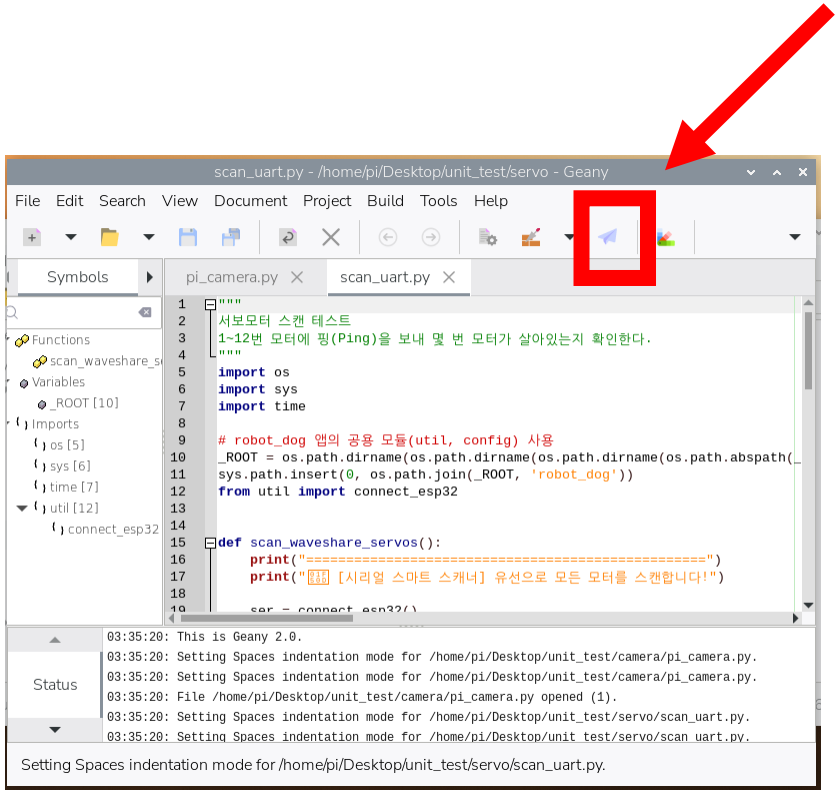




2. Servo test

First, Run Scan_uart.py, you can see that the ESP32 is successfully connected.

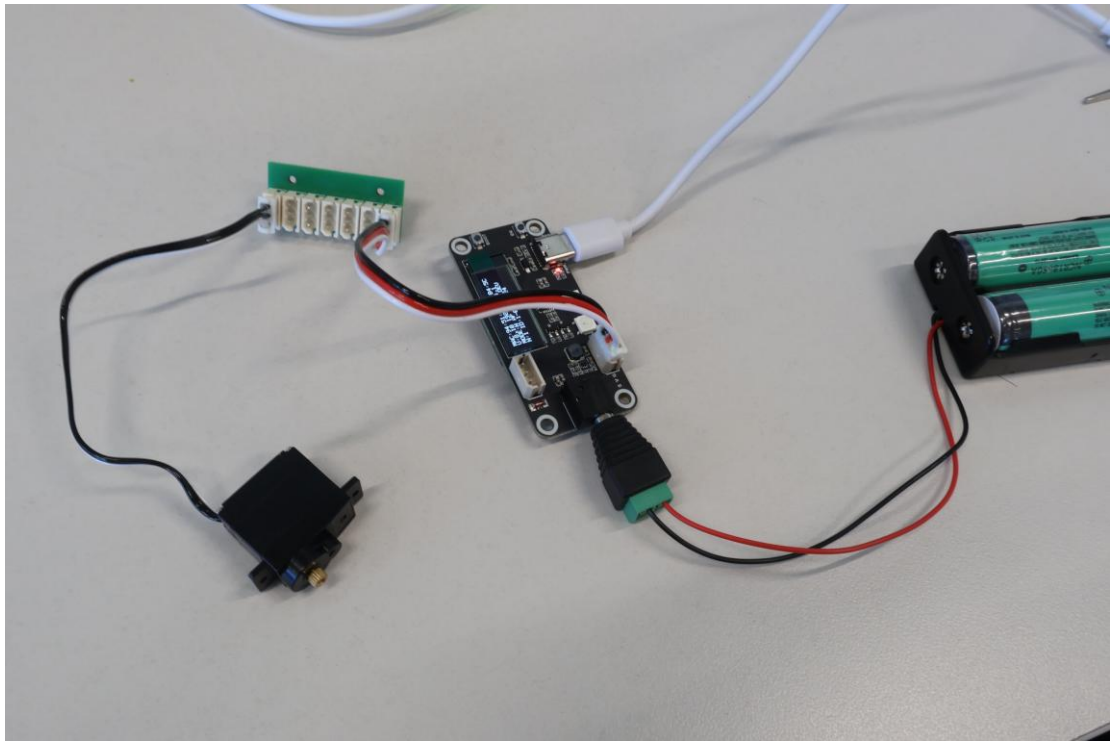




However, none of the servo motors are showing up. Why is this happening?

2. Servo test

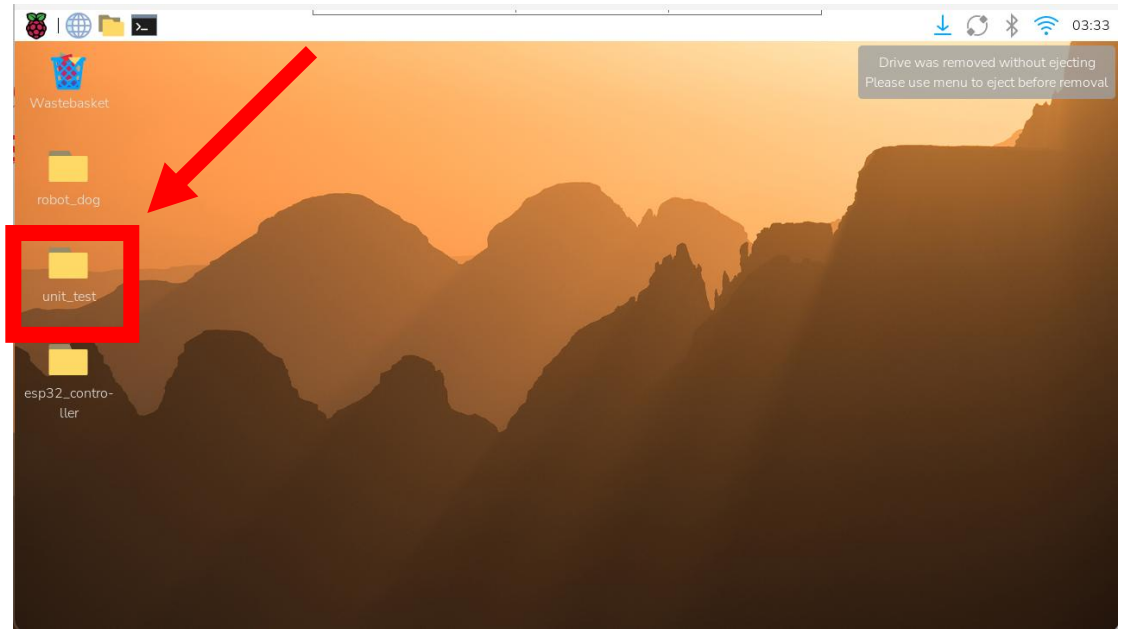
This is because the servos haven't been assigned IDs yet
Let's assign them IDs in numerical order

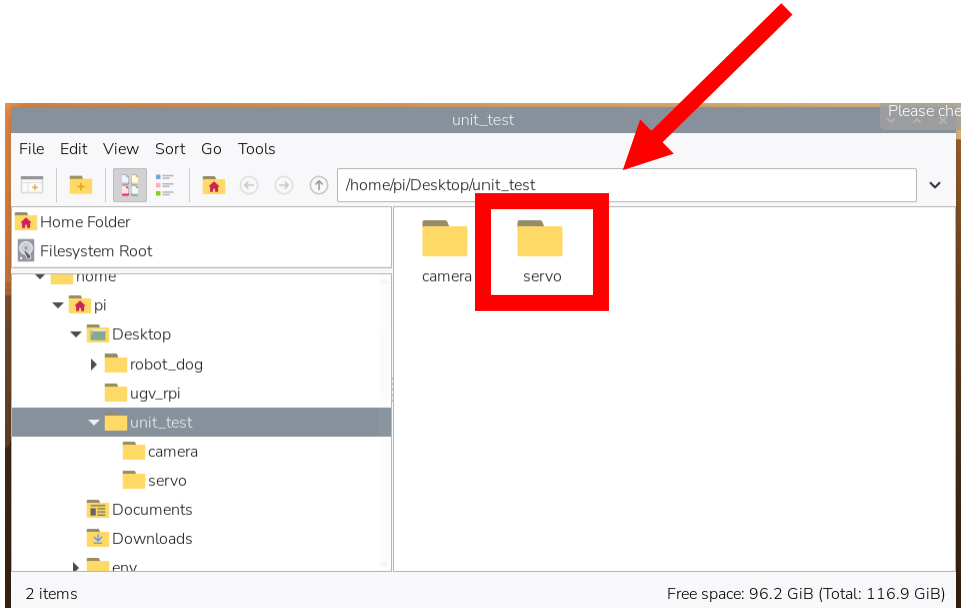


Connect the ESP to the Raspberry Pi.

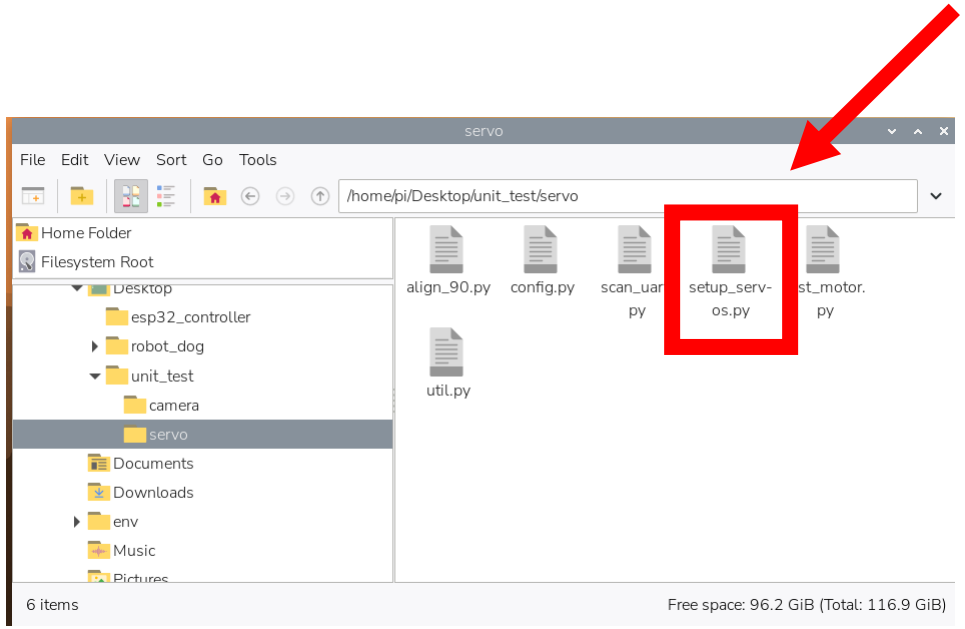
Only one servo motor should be connected when setting the IDs.

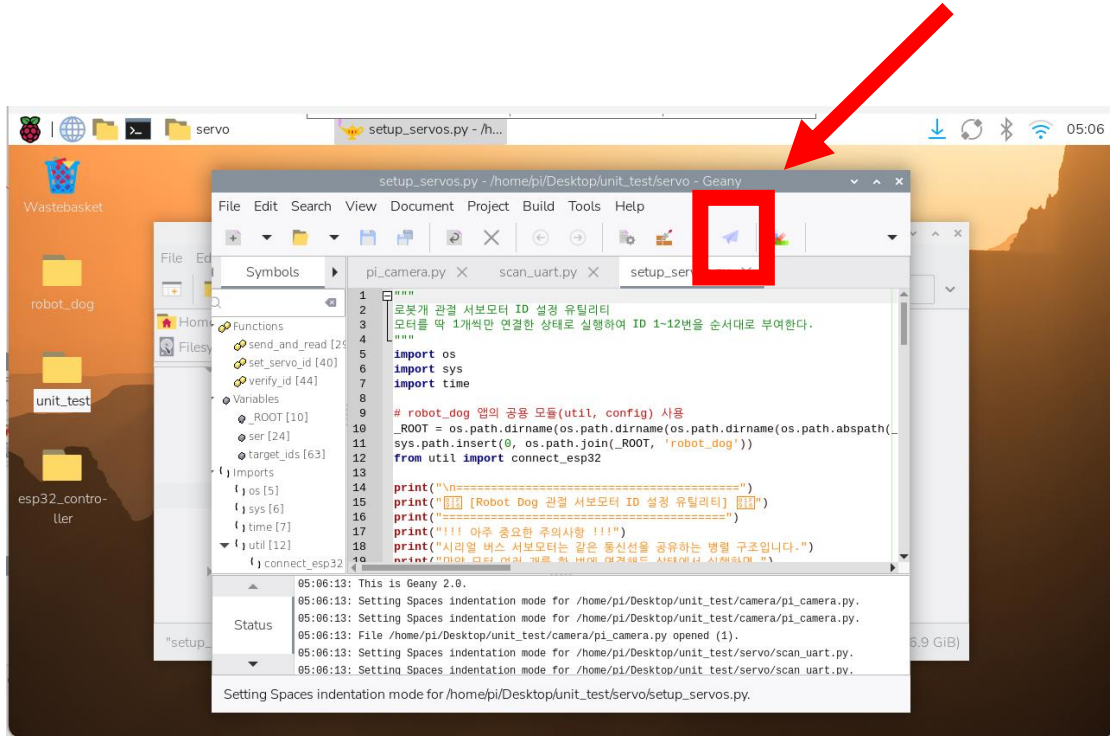
2. Servo test


03:33
Drive was removed without ejecting
Please use menu to eject before removal
Wastebasket
robot_dog
unit_test
esp32_contro-
ller


unit_test
File Edit View Sort Go Tools
/home/pi/Desktop/unit_test
Home Folder
Filesystem Root
home
pi
Desktop
robot_dog
ugv_rpi
unit_test
camera
servo
Documents
Downloads
env
camera
servo
2 items
Free space: 96.2 GiB (Total: 116.9 GiB)

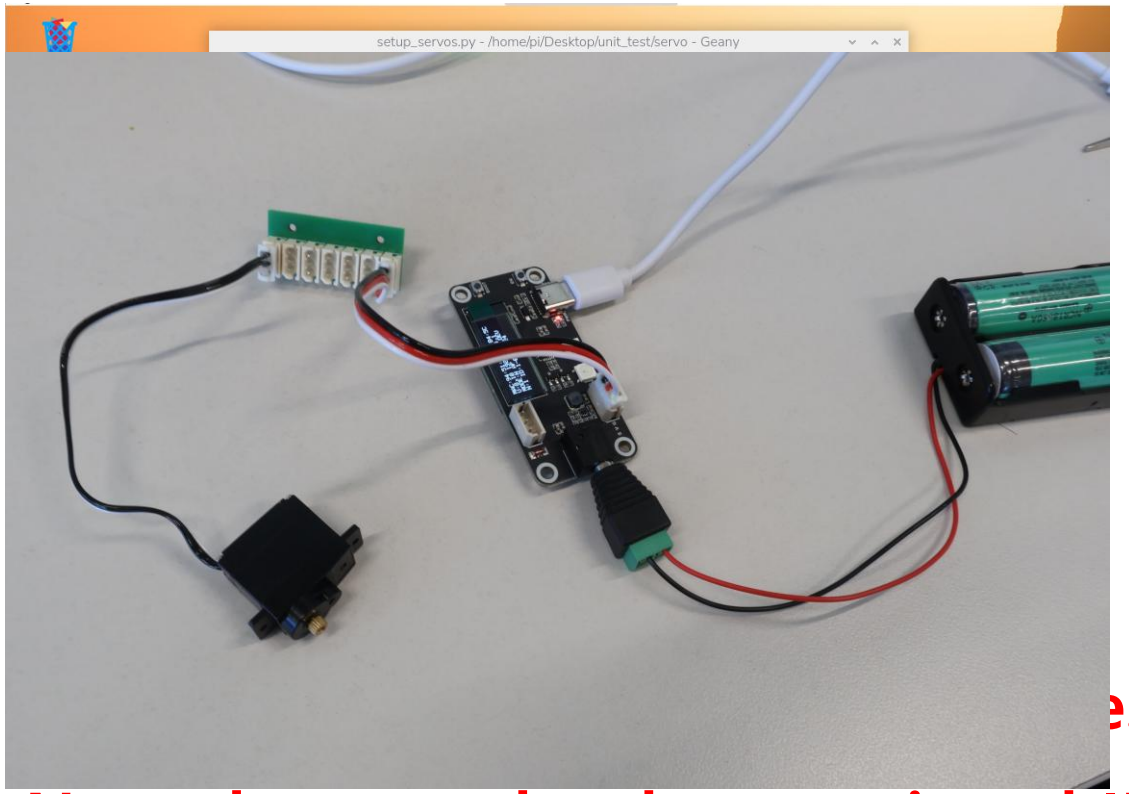
2. Servo test


servo
File Edit View Sort Go Tools
/home/pi/Desktop/unit_test/servo
Home Folder
Filesystem Root
Desktop
esp32_controller
robot_dog
unit_test
camera
servo
Documents
Downloads
env
Music
Pictures
align_90.py
config.py
scan_uar
py
setup_serv-
os.py
st_motor.
py
util.py
6 items
Free space: 96.2 GiB (Total: 116.9 GiB)


servo
setup_servos.py - /h...
05:06
Wastebasket
robot_dog
unit_test
esp32_contro-
ller
setup_servos.py - /home/pi/Desktop/unit_test/servo - Geany
File Edit Search View Document Project Build Tools Help
Symbols
pi_camera.py
scan_uart.py
setup_serv
Functions
send_and_read [29
set_servo_id [40]
verify_id [44]
Variables
_ROOT [10]
ser [24]
target_ids [63]
Imports
os [5]
sys [6]
time [7]
util [12]
connect_esp32
"""
로봇개 관절 서보모터 ID 설정 유틸리티
모터를 딱 1개씩만 연결한 상태로 실행하여 ID 1~12번을 순서대로 부여한다.
"""
import os
import sys
import time
robot_dog 앱의 공용 모듈(util, config) 사용
ROOT = os.path.dirname(os.path.dirname(os.path.dirname(os.path.abspath(
sys.path.insert(0, os.path.join(_ROOT, 'robot_dog'))
from util import connect_esp32
print("\n==")
print("[[Robot Dog 관절 서보모터 ID 설정 유틸리티]]")
print("==")
print("!!! 아주 중요한 주의사항 !!!")
print("시리얼 버스 서보모터는 같은 통신선을 공유하는 병렬 구조입니다.")
Status
05:06:13: This is Geany 2.0.
05:06:13: Setting Spaces indentation mode for /home/pi/Desktop/unit_test/camera/pi_camera.py.
05:06:13: Setting Spaces indentation mode for /home/pi/Desktop/unit_test/camera/pi_camera.py.
05:06:13: File /home/pi/Desktop/unit_test/camera/pi_camera.py opened (1).
05:06:13: Setting Spaces indentation mode for /home/pi/Desktop/unit_test/servo/scan_uart.py.
05:06:13: Setting Spaces indentation mode for /home/pi/Desktop/unit_test/servo/scan_uart.py.
Setting Spaces indentation mode for /home/pi/Desktop/unit_test/servo/setup_servos.py.

2. Servo test

Run the code with only one servo motor connected.

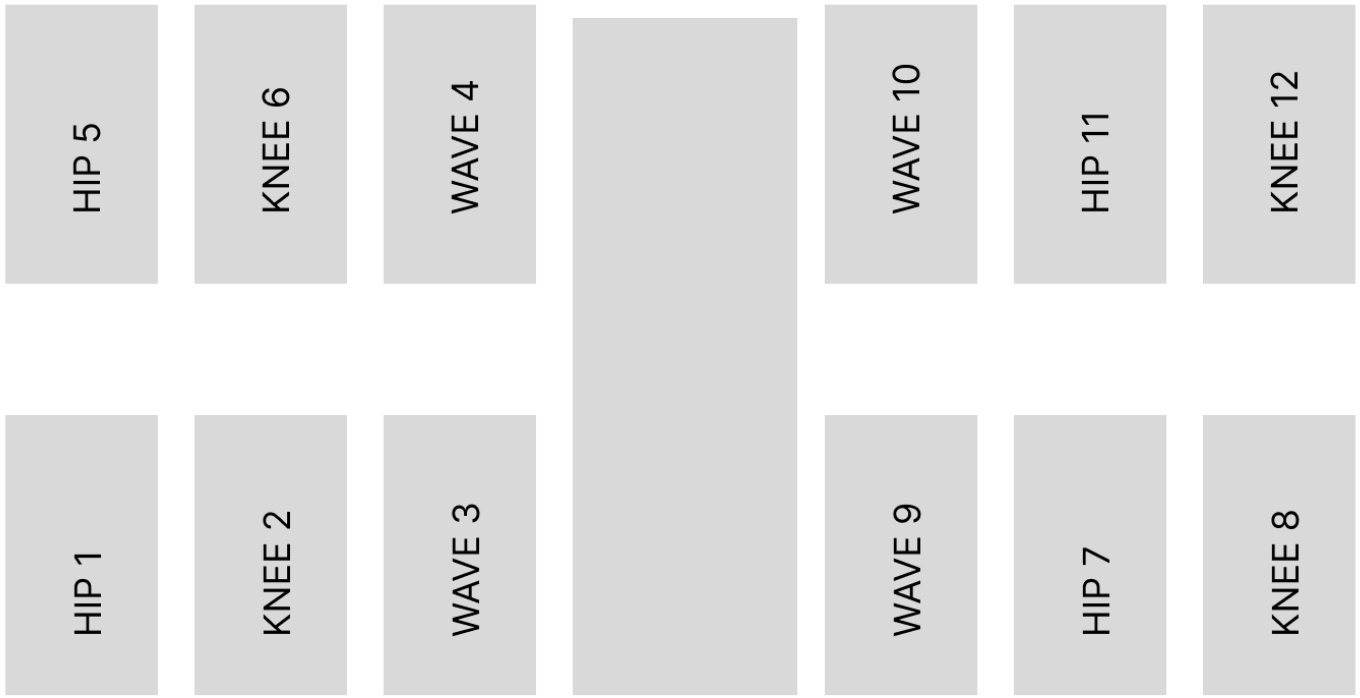


e.

Now, the servo has been assigned ID 1.

2. Servo test

Next, connect another motor and press Enter again to save it as ID 2.
Repeat this process up to ID 12.

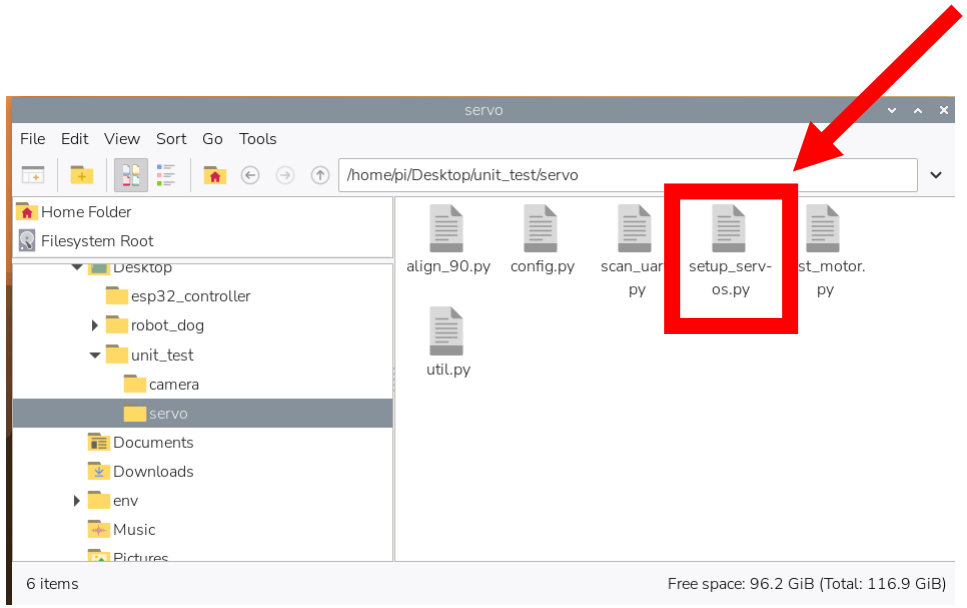


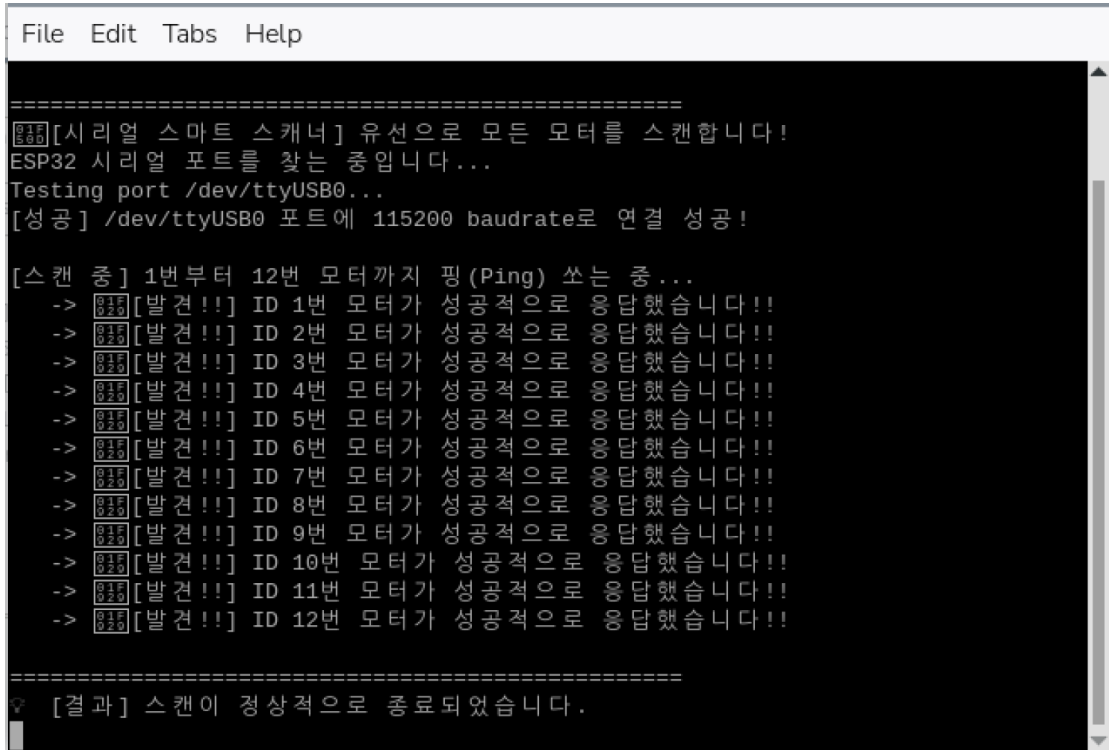
Camera

Please assign the IDs to the servos matching the positions and numbers shown in this diagram. Otherwise, it will not function properly.

2. Servo test

Once the setup is complete, run the scan_uart code again.

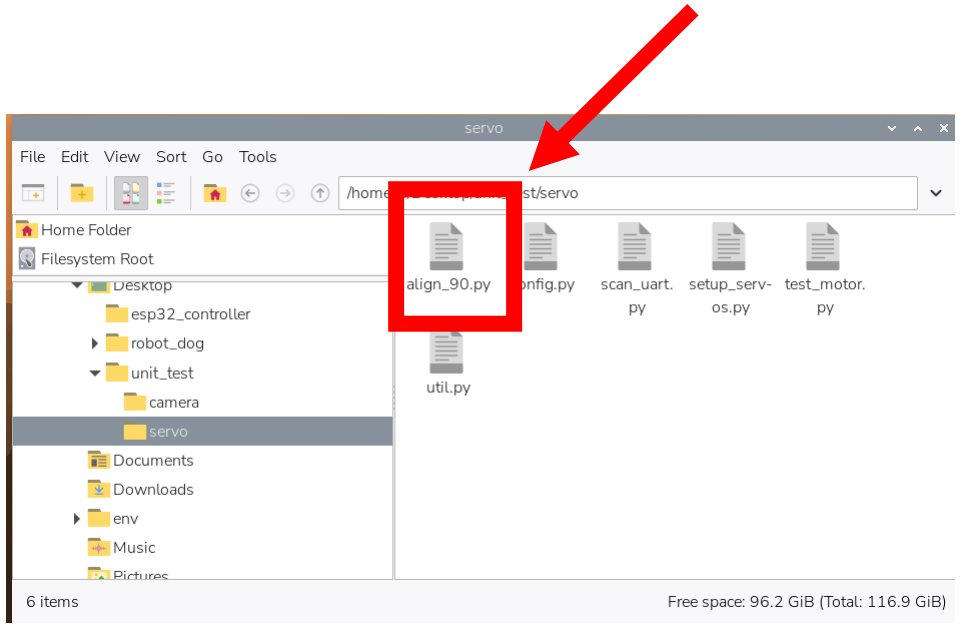




If you see this log, it is successful.

2. Servo test

Now, set all the servos to 90 degrees.
Please connected all servo once like this



Now all the servomotors are set to 90 degrees.
Please keep track of the number assigned to each servo.